

VALUE ADDED COURSE ON AI/ ML - TASK 6

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SUBJECT: ARTIFICIAL INTELLIGENCE

EXPERIMENT 6: Automated News Headline Categorizer using Multi-Class Text Classification

AIM:

To develop a multi-class news headline classification system using TF-IDF vectorization and Linear Support Vector Classifier (LinearSVC) to automatically categorize news into World, Sports, Business, and Sci/Tech.

SYSTEM REQUIREMENTS:

Hardware Requirements

- **Processor:** Intel Core i3 or equivalent (minimum)
- **RAM:** 4 GB or higher
- **Internet Connection:** Required (for loading raw dataset directly in Google Colab)

Software Requirements

- **Operating System:** Windows 10/11, macOS, or Linux
- **Platform:** Google Colab / Jupyter Notebook
- **Environment:** Python 3.x
- **Required Libraries:** pandas, numpy, scikit-learn

PROCEDURE:

1. **Dataset Acquisition:** Load the AG News dataset using Pandas from the provided URL.
2. **Category Mapping:** Map numerical class labels to four categories: World, Sports, Business, and Sci/Tech.
3. **Text Preprocessing:** Remove HTML tags, punctuation, special characters, and convert text to lowercase.
4. **Feature Combination:** Combine news headlines and descriptions into a single text column.
5. **TF-IDF Vectorization:** Convert text into numerical features using unigrams and bigrams while removing English stopwords.
6. **Train-Test Split:** Divide the dataset into 80% training data and 20% testing data using stratified sampling.
7. **Model Training:** Train the LinearSVC model using the extracted TF-IDF features.
8. **Model Evaluation:** Evaluate the model using accuracy, confusion matrix, precision, recall, and F1-score.

9. **Prediction:** Test the trained model with new headlines and display their predicted news categories.

PROGRAM:

```
# ============================================================================
# Experiment 6: Automated News Headline Categorizer
# Dataset: AG News Benchmark Dataset (Multi-Class Text Classification)
# ============================================================================

import pandas as pd

import numpy as np

import re

from sklearn.model_selection import train_test_split

from sklearn.feature_extraction.text import TfidfVectorizer

from sklearn.svm import LinearSVC

from sklearn.metrics import accuracy_score, confusion_matrix, classification_report


# Step 1: Load Benchmark Dataset (AG News CSV)

DATA_URL =
"https://raw.githubusercontent.com/mhjabreel/CharSCNN/master/data/ag_news_csv/train.csv"

df = pd.read_csv(DATA_URL, header=None, names=['ClassIndex', 'Title',
'Description']).head(10000)


# Step 2: Map Target Class Labels

# Class 1: World | Class 2: Sports | Class 3: Business | Class 4: Sci/Tech

category_map = {1: 'World', 2: 'Sports', 3: 'Business', 4: 'Sci/Tech'}

df['Category'] = df['ClassIndex'].map(category_map)


print("--- Step 1: Dataset Overview ---")

print(f"Total News Samples Loaded: {df.shape[0]}\n")

print(df[['Category', 'Title']].head())

print("\n" + "="*60 + "\n")
```

```
# Step 3: Text Preprocessing & Feature Synthesis
def clean_headline(text):
    text = re.sub(r'<.*?>', '', text)           # Remove HTML tags
    text = re.sub(r'[^a-zA-Z\s]', '', text)      # Remove special characters & numbers
    text = text.lower().strip()                 # Lowercase & trim
    return text

# Combine Title and Description into a single rich text column
df['full_text'] = (df['Title'] + " " + df['Description']).apply(clean_headline)

X = df['full_text']
y = df['Category']

# Step 4: Text Feature Vectorization (TF-IDF with Unigrams and Bigrams)
tfidf = TfidfVectorizer(ngram_range=(1, 2), max_features=10000, stop_words='english')
X_tfidf = tfidf.fit_transform(X)

print(f"TF-IDF Feature Matrix Shape: {X_tfidf.shape[0]} rows x {X_tfidf.shape[1]} n-gram features")
print("="*60 + "\n")

# Step 5: Train-Test Split (80% Train, 20% Test)
X_train, X_test, y_train, y_test = train_test_split(
    X_tfidf, y, test_size=0.2, random_state=42, stratify=y
)

# Step 6: Model Training (Linear Support Vector Classifier)
model = LinearSVC(C=1.0, max_iter=1000)
model.fit(X_train, y_train)
```

```
# Step 7: Prediction & Evaluation
y_pred = model.predict(X_test)

acc = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred, labels=['World', 'Sports', 'Business', 'Sci/Tech'])
report = classification_report(y_test, y_pred)

print("--- Step 2: Model Performance Metrics ---")
print(f"Multi-Class Classification Accuracy: {acc * 100:.2f}%\n")

print("Confusion Matrix (World | Sports | Business | Sci/Tech):")
print(cm)
print("\nDetailed Multi-Class Classification Report:")
print(report)

# Step 8: Custom Inference Function for Unseen Headlines
def categorize_headline(headline):
    cleaned = clean_headline(headline)
    vec = tfidf.transform([cleaned])
    prediction = model.predict(vec)[0]

    print(f"\nHeadline: \"{headline}\"")
    print(f"Predicted Category: [{prediction.upper()}]")

print("="*60)
print("--- Step 3: Real-Time Headline Categorization Test ---")
categorize_headline("Federal Reserve hints at interest rate cuts as inflation stabilizes.")
categorize_headline("NASA's James Webb Space Telescope discovers galaxy in deep space.")
categorize_headline("Real Madrid wins Champions League thriller with late stoppage-time goal.")
categorize_headline("United Nations holds emergency summit over ongoing diplomatic conflict.")
print("="*60)
```

OUTPUT:

```
Total News Samples: 10000
   ClassIndex                                             Title
0           3  Wall St. Bears Claw Back Into the Black (Reuters)
1           3  Carlyle Looks Toward Commercial Aerospace (Reu...
2           3    Oil and Economy Cloud Stocks' Outlook (Reuters)
3           3  Iraq Halts Oil Exports from Main Southern Pipe...
4           3  Oil prices soar to all-time record, posing new...
TF-IDF Matrix Shape: (10000, 10000)

Accuracy: 89.45 %

Confusion Matrix:
[[436  27  28  14]
 [  8 459   0   1]
 [ 20   3 433  39]
 [ 19   7  45 461]]

Classification Report:
              precision    recall  f1-score   support

    Business       0.86      0.87      0.87       495
    Sci/Tech       0.90      0.87      0.88       532
      Sports       0.93      0.98      0.95       468
       World       0.90      0.86      0.88       505

    accuracy                           0.89      2000
   macro avg       0.89      0.90      0.90      2000
weighted avg       0.89      0.89      0.89      2000


--- News Categorization Tests ---

Headline: Federal Reserve hints at interest rate cuts as inflation stabilizes.
Predicted Category: Business

Headline: NASA's James Webb Space Telescope discovers galaxy in deep space.
Predicted Category: Sci/Tech

Headline: Real Madrid wins Champions League with late goal.
Predicted Category: Sports

Headline: United Nations holds emergency summit over diplomatic conflict.
Predicted Category: World
```

RESULT:

Thus, the automated news headline categorization system was successfully developed using TF-IDF vectorization and LinearSVC. The model classifies news into World, Sports, Business, and Sci/Tech categories and evaluates its performance using accuracy, precision, recall, and F1-score.